LESSON 01 / 10

One distance ratio creates three curves

Fix a focus F, a directrix ℓ not passing through F, and $e > 0$. A point P lies on the corresponding conic when

$$PF = e\,\mathrm{dist}(P, \ell).$$

The ratio is the **eccentricity**: $0 < e < 1$ gives an ellipse, $e = 1$ a parabola, and $e > 1$ a hyperbola. A circle is the limiting ellipse with $e = 0$ and its directrix at infinity; putting $e = 0$ into the finite-line equation instead collapses the locus to the focus itself.

KEY RESULT

Classify before calculating

The value of e names the curve. Do not apply an ellipse relation such as $b^2 = a^2 - c^2$ to a hyperbola, where $b^2 = c^2 - a^2$. First decide the curve, then choose its formula family.

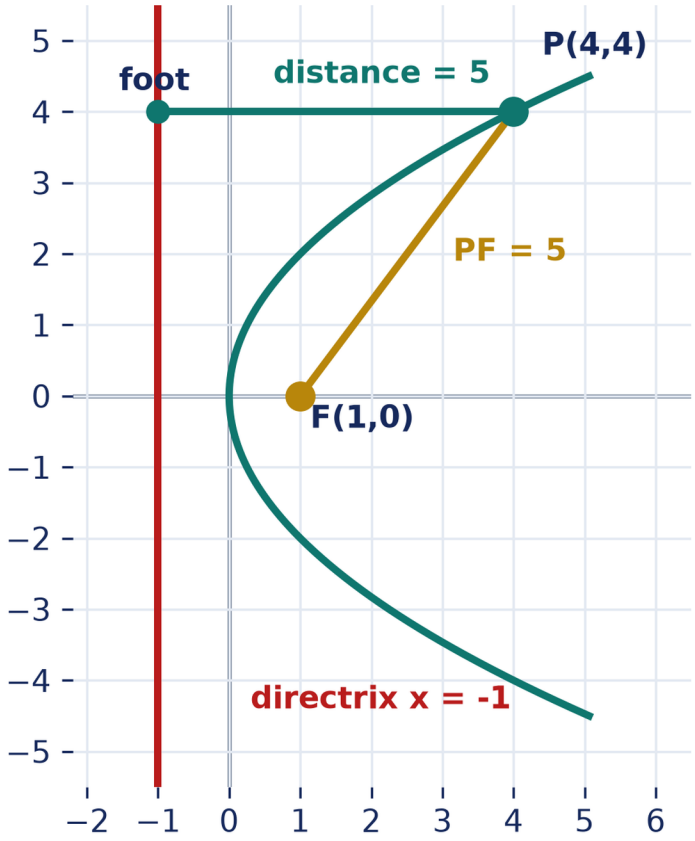
The parabola is one squaring away

For $F = (a, 0)$ and directrix $x = -a$, the vertex is midway between them. At $P = (x, y)$,

$$(x - a)^2 + y^2 = (x + a)^2 \quad \Longrightarrow \quad y^2 = 4ax.$$

Here $a > 0$, the focus is $(a, 0)$, the axis is $y = 0$, and the latus rectum has endpoints $(a, \pm 2a)$ and length $4a$. On the curve, $PF = x + a$: a focal distance becomes a coordinate, not a square root.

On y^2 = 4x, both distances at P(4,4) equal 5



READ THE FIGURE
The curve is y^2 = 4x, so a = 1. PF = sqrt(3^2 + 4^2) = 5 and the distance to x = -1 is also 5.

Central conics use two foci

For an ellipse with foci $(\pm c, 0)$ and $a > c$,

$$PF_1 + PF_2 = 2a, \qquad b^2 = a^2 - c^2, \qquad \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Its eccentricity is $e = c/a < 1$, its directrices are $x = \pm a/e$, and its focal distances are $a \pm ex$. For a hyperbola with $c > a$,

$$|PF_1 - PF_2| = 2a, \qquad b^2 = c^2 - a^2, \qquad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1.$$

Now $e = c/a > 1$, the directrices remain $x = \pm a/e$, and the absolute difference selects both branches.

The anatomy dictionary

Curve	Distance law	Parameter relation	Latus rectum
Parabola	PF = line distance	a = focal length	4a
Ellipse	sum = 2a	c^2 = a^2 - b^2	2b^2/a
Hyperbola	|difference| = 2a	c^2 = a^2 + b^2	2b^2/a

READ THE FIGURE

For both central conics c = ae and the directrices are x = +/- a/e. The parameter relation carries the crucial sign.

WORKED EXAMPLE

Recover an ellipse from focus–directrix data

A point moves so that its distance from $(2,0)$ is $2/3$ of its distance from $x=9/2$. Thus $e=2/3<1$. Squaring gives

$$(x-2)^2+y^2=\tfrac{4}{9}\left(x-\tfrac{9}{2}\right)^2 \quad\Longrightarrow\quad 5x^2+9y^2=45.$$

Hence the locus is $x^2/9+y^2/5=1$. As a check, $a=3$, $ae=2$, and $a/e=9/2$, exactly the supplied focus and directrix.

COMMON TRAP

Degeneracy is a separate question

A second-degree equation can represent a point, a line, two lines, or no real points. The simple e-classification assumes a genuine nondegenerate conic with the focus off the directrix. Always check degenerate data explicitly rather than forcing it into a standard formula.

LESSON 02 / 10

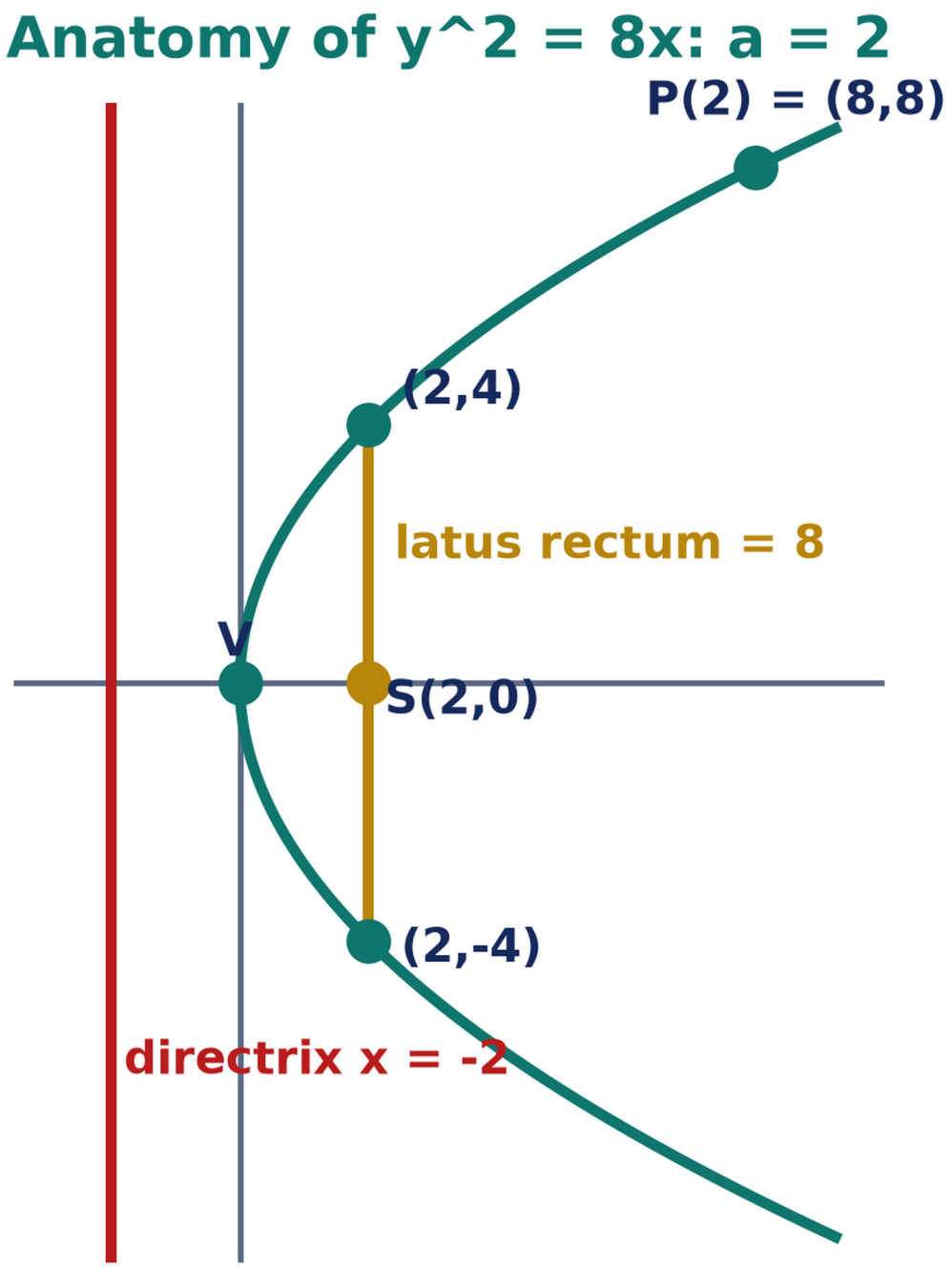
Read the anatomy before starting algebra

For $y^2 = 4ax$ with $a > 0$: vertex $(0, 0)$, focus $S(a, 0)$, directrix $x = -a$, axis $y = 0$, and latus rectum endpoints $(a, \pm 2a)$. At $P(x, y)$ on the curve, $SP = x + a$. The reflected forms are $y^2 = -4ax$, $x^2 = 4ay$, and $x^2 = -4ay$; a vertex shift (h, k) gives $(y - k)^2 = 4a(x - h)$ and its analogues.

COMMON TRAP

Match $4a$, not a

For $y^2 = 16x$, $4a = 16$, so $a = 4$. The focus is not $(16, 0)$. Also reverse the focal-distance sign when the parabola opens left: always derive it from the corresponding directrix rather than memorizing blindly.



READ THE FIGURE
At P(t) = (at^2, 2at), t = 2 gives (8,8). Its focal distance is x + a = 10 = a(1 + t^2).

The parameter turns coordinates into one variable

Every point of $y^2 = 4ax$ is

$$P(t) = (at^2, 2at), \qquad t \in \mathbb{R}.$$

Immediately $SP = a(1 + t^2)$. The chord through $P(t_1)$ and $P(t_2)$ is

$$(t_1 + t_2)y = 2x + 2at_1t_2.$$

Its slope is $2/(t_1 + t_2)$ when $t_1 + t_2 \neq 0$. Substituting the focus $(a, 0)$ into the chord equation proves the essential focal-chord test

$$t_1 t_2 = -1.$$

Thus one end $P(t)$ determines the other $P(-1/t)$. Their focal distances are $a(1+t^2)$ and $a(1+1/t^2)$, so the full chord length is

$$a\left(2 + t^2 + \tfrac{1}{t^2}\right) = a\left(t + \tfrac{1}{t}\right)^2.$$

The minimum is $4a$ at $t = \pm 1$, the latus rectum.

SHORT DERIVATION

The semi-latus rectum is a harmonic mean

If a focal chord is divided by the focus into lengths u and v, then

$$\tfrac{1}{u} + \tfrac{1}{v} = \tfrac{1}{a} = \tfrac{2}{2a}.$$

Therefore the harmonic mean $2/(1/u + 1/v)$ is $2a$, the semi-latus rectum. For an ellipse the corresponding constant is b^2/a. Hyperbola focal chords require branch-aware signs, so this unsigned formula must not be transferred automatically.

WORKED EXAMPLE

A focal chord with exact endpoints

For $y^2 = 8x$, $a = 2$. Take $t = 2$, so one endpoint is $P(2) = (8, 8)$. The other parameter is $-1/2$, giving

$$P\left(-\tfrac{1}{2}\right) = \left(\tfrac{1}{2}, -2\right).$$

The two focal parts are 10 and $5/2$, so the chord length is $25/2$. Also $1/10 + 2/5 = 1/2 = 1/a$, confirming the harmonic-mean relation.

KEY RESULT

When to choose parameters

Use t for chords through named points, focal chords, tangent intersections, and normals. Use direct (x, y) substitution for a single focal distance or a quick membership check. The parameter is not an angle.

LESSON 03 / 10

Two lines at $P(t)$ do most of the work

At $P(t) = (at^2, 2at)$ on $y^2 = 4ax$,

$$\text{tangent: } ty = x + at^2, \qquad \text{normal: } y = -tx + 2at + at^3.$$

For a finite tangent slope $m = 1/t \neq 0$,

$$y = mx + \frac{a}{m}, \qquad \text{contact } \left(\frac{a}{m^2}, \frac{2a}{m}\right).$$

For a normal of slope $m = -t$, $y = mx - 2am - am^3$. Tangents at t_1, t_2 meet at

$$T = (at_1t_2, a(t_1 + t_2)).$$

Consequently they are perpendicular exactly when their slopes multiply to -1, namely $t_1t_2 = -1$; their intersection then lies on $x = -a$, the directrix.

WORKED EXAMPLE

The focal-chord and directrix facts agree

Use the focal chord of $y^2 = 8x$ with parameters 2 and $-1/2$. Here $a = 2$ and $t_1t_2 = -1$. The tangents meet at

$$T = \left(2 \cdot (-1), 2\left(2 - \tfrac{1}{2}\right)\right) = (-2, 3),$$

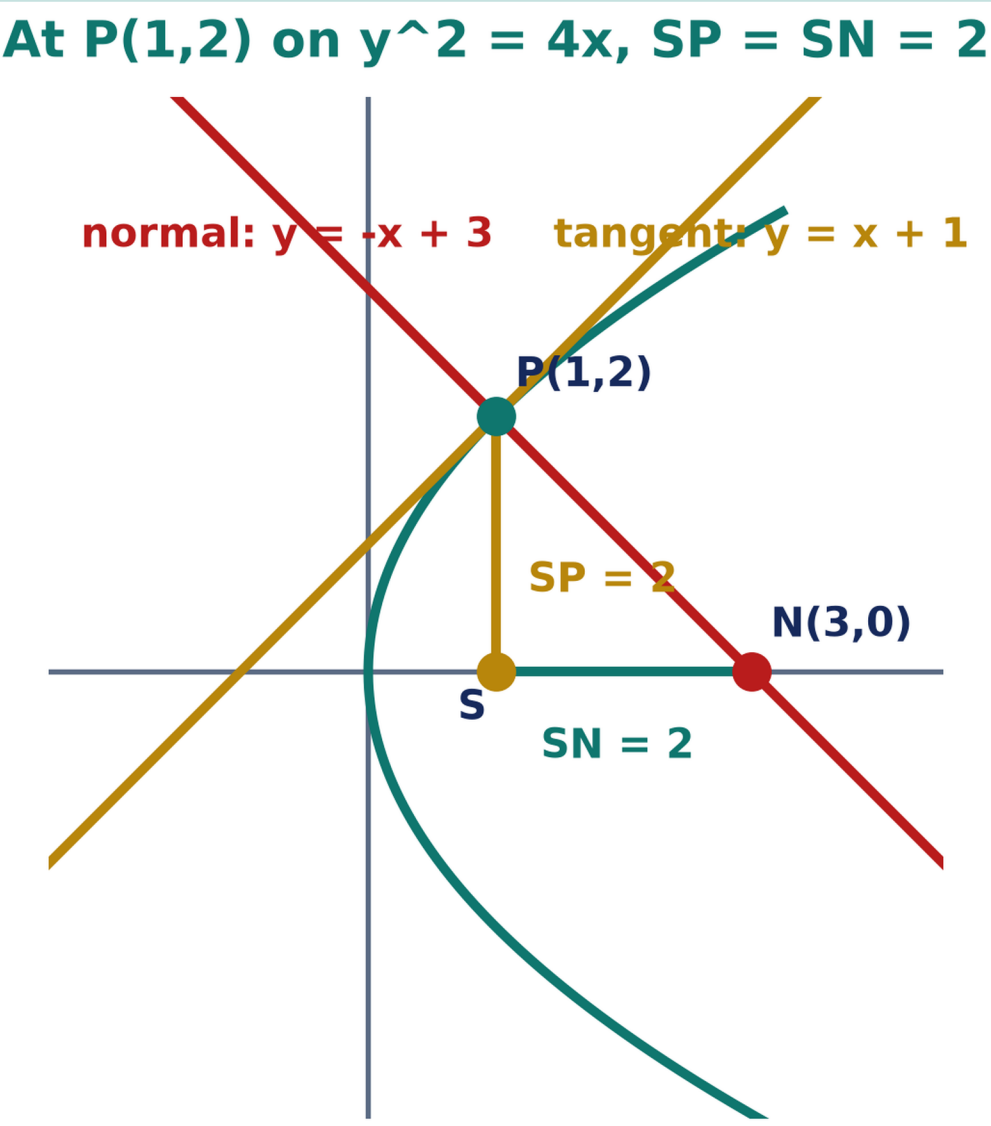
which really is on the directrix $x = -2$. Their slopes $1/2$ and -2 multiply to -1. Notice that the parameters, not merely the endpoint coordinates, are needed for this conclusion.

Axis intercepts give quick geometric quantities

The tangent at $P(t)$ meets the axis at $(-at^2, 0)$. For $t \neq 0$, the normal meets it at $N = (at^2 + 2a, 0)$, and the subnormal is constantly $2a$; at $t = 0$, the normal is the axis itself. At the focus $S(a, 0)$,

$$SN = a(1 + t^2) = SP.$$

So triangle SPN is isosceles. The normal bisects the angle between the focal ray and the axis direction; this proves the reflection property: rays parallel to the axis reflect through the focus, and rays from the focus emerge parallel to the axis.



READ THE FIGURE
Here $a = 1$ and $t = 1$. The tangent and normal are perpendicular, while equal lengths SP and SN explain the equal-angle reflection property.

Normals from an external point form a cubic

A normal of slope m passes through (h, k) precisely when

$$am^3 + (2a - h)m + k = 0.$$

There are at most three real normals. Vieta also gives $m_1 + m_2 + m_3 = 0$ for the cubic roots; in particular, when three real normals exist, their slopes sum to zero. A cubic root count must still be checked—having three algebraic roots does not say all are real and distinct.

WORKED EXAMPLE

All normals from $(6,0)$ ***to*** $y^2 = 4x$

With $a = 1$, the cubic becomes

$$m^3 - 4m = m(m-2)(m+2) = 0.$$

The slopes are $0, 2, -2$. Since the contact parameter is $t = -m$, the feet are $(0,0), (4,-4), (4,4)$, respectively. The three normal lines are $y = 0$, $y = 2x - 12$, and $y = -2x + 12$; all pass through $(6,0)$.

Tangency is a repeated-root condition

Substituting $y = mx + c$ into $y^2 = 4ax$ gives

$$m^2x^2 + (2mc - 4a)x + c^2 = 0.$$

A tangent has a repeated root, so its discriminant is zero and $c = a/m$ for $m \neq 0$. This is the derivation behind the slope form, and it also supplies a quick independent check: substitution at the stated contact must produce a perfect square.

WORKED EXAMPLE

Two tangents from a point on the directrix

For $y^2 = 8x$, take the external point $(-2,0)$ on the directrix. A tangent $y = mx + 2/m$ passes through it when

$$0 = -2m + \tfrac{2}{m} \quad \Longrightarrow \quad m^2 = 1.$$

The tangents are $y = x + 2$ and $y = -x - 2$, with contacts $(2,4)$ and $(2,-4)$. Their slopes are perpendicular, and the contact chord is the latus rectum $x = 2$.

COMMON TRAP

Do not invent a directrix

The intersection formula gives $x = at_1t_2$. Only when $t_1t_2 = -1$ does this become $x = -a$. Two arbitrary tangents need not be perpendicular and need not meet on the directrix. Verify the parameter product before quoting the focal-chord theorem.

LESSON 04 / 10

A constant sum determines the ellipse

For foci $F_1 = (-c, 0)$ and $F_2 = (c, 0)$, require $PF_1 + PF_2 = 2a$ with $a > c$. Setting $b^2 = a^2 - c^2$ yields

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \qquad a > b > 0.$$

The vertices are $(\pm a, 0)$, covertices $(0, \pm b)$, foci $(\pm ae, 0)$ where $e = c/a$, directrices $x = \pm a/e$, and latus rectum length $2b^2/a$. The circle is the special case $a = b$ and $e = 0$.

SHORT DERIVATION

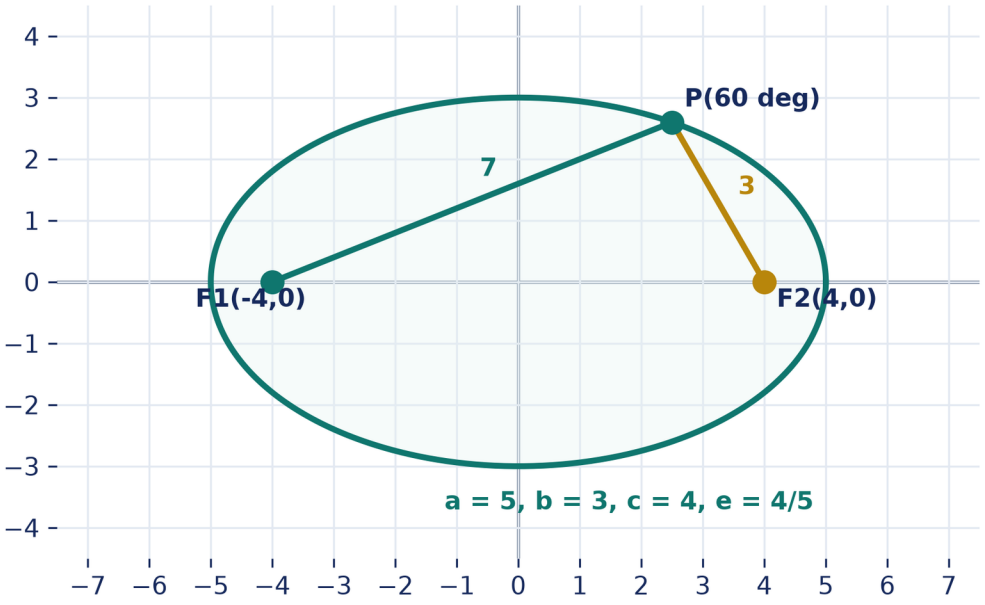
Focal distances without square roots

For $P = (x, y)$ on the ellipse, the right focus-directrix pair gives

$$PF_{\text{right}} = e\left(\frac{a}{e} - x\right) = a - ex,$$

and the left focal distance is $a + ex$. Their sum is $2a$. These linear formulas are usually faster than recomputing two radicals.

On x^2/25 + y^2/9 = 1, the focal distances are 3 and 7



READ THE FIGURE

P = (5/2, 3 sqrt(3)/2). Since x = 5/2, a - ex = 3 and a + ex = 7. Their sum is 10 = 2a.

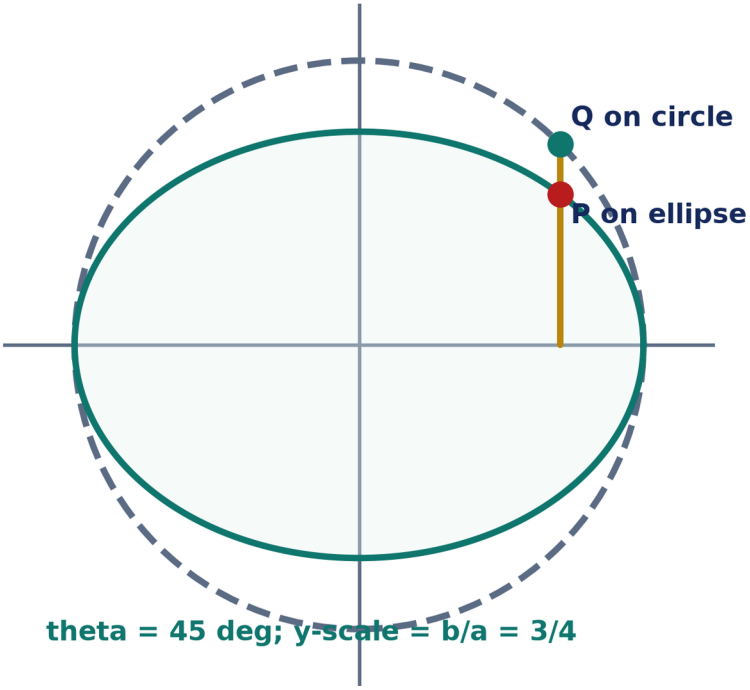
The eccentric angle belongs to the auxiliary circle

The auxiliary circle is $x^2 + y^2 = a^2$. If $Q = (a\cos\theta, a\sin\theta)$ lies on it, squashing vertically by b/a gives the ellipse point

$$P(\theta) = (a\cos\theta, b\sin\theta).$$

Thus θ is the angle of Q on the circle, not generally the polar angle of P. This parametrization is the fastest route to tangent and normal formulas.

The ellipse is the auxiliary circle squashed vertically



READ THE FIGURE
Q = (2 sqrt(2), 2 sqrt(2)) maps to P = (2 sqrt(2), 3/sqrt(2)). The x-coordinate is unchanged; the y-coordinate is multiplied by 3/4.

WORKED EXAMPLE

Reconstruct an ellipse from foci and one point

An ellipse centred at the origin has foci $(\pm 2, 0)$ and passes through $(2, 3)$. The two distances at this point are 3 and 5, so $2a = 8$ and $a = 4$. Since $c = 2$,

$$b^2 = a^2 - c^2 = 16 - 4 = 12.$$

Therefore the ellipse is $x^2/16 + y^2/12 = 1$, with $e = c/a = 1/2$. Substitution at $(2, 3)$ gives $1/4 + 3/4 = 1$.

COMMON TRAP

Major-axis convention matters

The formulas above use $a > b$ and foci on the x-axis. If the larger denominator is under y^2, the major axis is vertical: swap the coordinate roles before naming foci and directrices. Do not force every ellipse into horizontal notation.

LESSON 05 / 10

Tangent and normal in the form that fits the data

At (x_1, y_1) on $x^2/a^2 + y^2/b^2 = 1$, and at eccentric angle θ,

$$\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1, \qquad \frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1.$$

A finite-slope tangent has the two forms

$$y = mx \pm \sqrt{a^2m^2 + b^2}.$$

The normal at eccentric angle θ is

$$ax\sin\theta - by\cos\theta = (a^2 - b^2)\sin\theta\cos\theta.$$

Point-form equations include axis tangents and normals that a finite-slope formula can miss.

WORKED EXAMPLE

Tangent at $\theta = 45^\circ$ on $x^2/16 + y^2/9 = 1$

The contact point is $P = (2\sqrt{2}, 3/\sqrt{2})$. The tangent is

$$\frac{x}{4\sqrt{2}} + \frac{y}{3\sqrt{2}} = 1 \quad\Longleftrightarrow\quad 3x + 4y = 12\sqrt{2}.$$

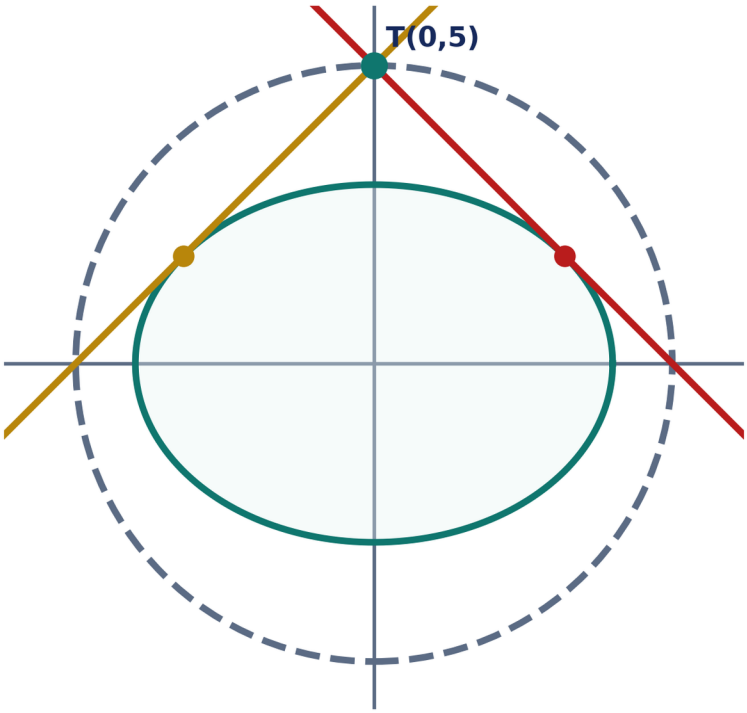
The normal simplifies to $4x - 3y = 7\sqrt{2}/2$. Their slopes are $-3/4$ and $4/3$, whose product is -1.

Position and perpendicular tangents

For $S = x^2/a^2 + y^2/b^2 - 1$, evaluating $S_1 = S(x_1, y_1)$ gives inside when $S_1 < 0$, on when $S_1 = 0$, and outside when $S_1 > 0$. The locus of intersections of perpendicular tangents is the **director circle**

$$x^2 + y^2 = a^2 + b^2.$$

Perpendicular tangents meet on the director circle



READ THE FIGURE
For a = 4 and b = 3, slope +/-1 tangents have intercept +/-5. The displayed perpendicular pair meets at (0,5), on x^2 + y^2 = 25.

A focal chord has a constant reciprocal sum

If a focus divides a focal chord into lengths u, v, then

$$\frac{1}{u}+\frac{1}{v}=\frac{2a}{b^2}.$$

Equivalently, the harmonic mean of u, v is the semi-latus rectum b^2/a. This is a relation between the two parts, not a claim that each part equals b^2/a.

WORKED EXAMPLE

An exact nonvertical focal chord

On $x^2/25+y^2/9=1$, take $P=(5/2, 3\sqrt{3}/2)$ and the right focus $F=(4,0)$. The line PF meets the ellipse again at

$$Q=\left(\frac{65}{14}, -\frac{9\sqrt{3}}{14}\right).$$

The two segment lengths are $PF=3$ and $QF=9/7$. Hence

$$\frac{1}{3}+\frac{7}{9}=\frac{10}{9}=\frac{2a}{b^2}, \qquad \frac{2}{1/3+7/9}=\frac{9}{5}=\frac{b^2}{a}.$$

Both endpoints satisfy the ellipse equation, and Q lies inside $|x|\le 5$ as it must.

Reflection explains the foci

At P, the tangent makes equal angles with the two focal segments; equivalently, the normal internally bisects the focal angle. Therefore a ray from one focus reflects toward the other. This follows geometrically from the constant sum $PF_1 + PF_2 = 2a$ and the stationary-path property of a tangent.

SHORT DERIVATION

The director circle from two tangent slopes

Let a tangent through $T = (h, k)$ have finite slope m. Since its intercept is $k - mh$, the ellipse tangency condition gives

$$(k - mh)^2 = a^2m^2 + b^2,$$

or

$$(h^2 - a^2)m^2 - 2hk\,m + (k^2 - b^2) = 0.$$

The two tangent slopes have product $(k^2 - b^2)/(h^2 - a^2)$. Perpendicularity makes this product -1, hence $h^2 + k^2 = a^2 + b^2$. Axis-aligned exceptional cases can be checked directly and lie on the same circle.

COMMON TRAP

A harmonic mean is not an arithmetic mean

The relation is $1/u + 1/v = 2a/b^2$. Do not replace it by $(u + v)/2 = b^2/a$, and do not accept a numerical endpoint without checking that it lies on the ellipse. Curve membership is the fastest way to catch a false focal-chord example.

LESSON 06 / 10

A constant absolute difference gives two branches

For foci $(\pm c, 0)$ and $c > a$, the condition $|PF_1 - PF_2| = 2a$ gives

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1, \qquad b^2 = c^2 - a^2.$$

The vertices are $(\pm a, 0)$, foci $(\pm ae, 0)$, $e = c/a > 1$, directrices $x = \pm a/e$, and latus rectum length $2b^2/a$. On the right branch, the near/right and far/left focal distances are $ex - a$ and $ex + a$; mirror the signs on the left branch.

COMMON TRAP

The absolute value matters

A fixed signed difference selects only one branch. The full hyperbola uses the absolute difference. Also $b^2 = c^2 - a^2$, not $a^2 - c^2$; now $c > a$, so the foci lie outside the vertices.

Asymptotes are the straight skeleton

Dividing the equation far from the centre shows that the curve approaches

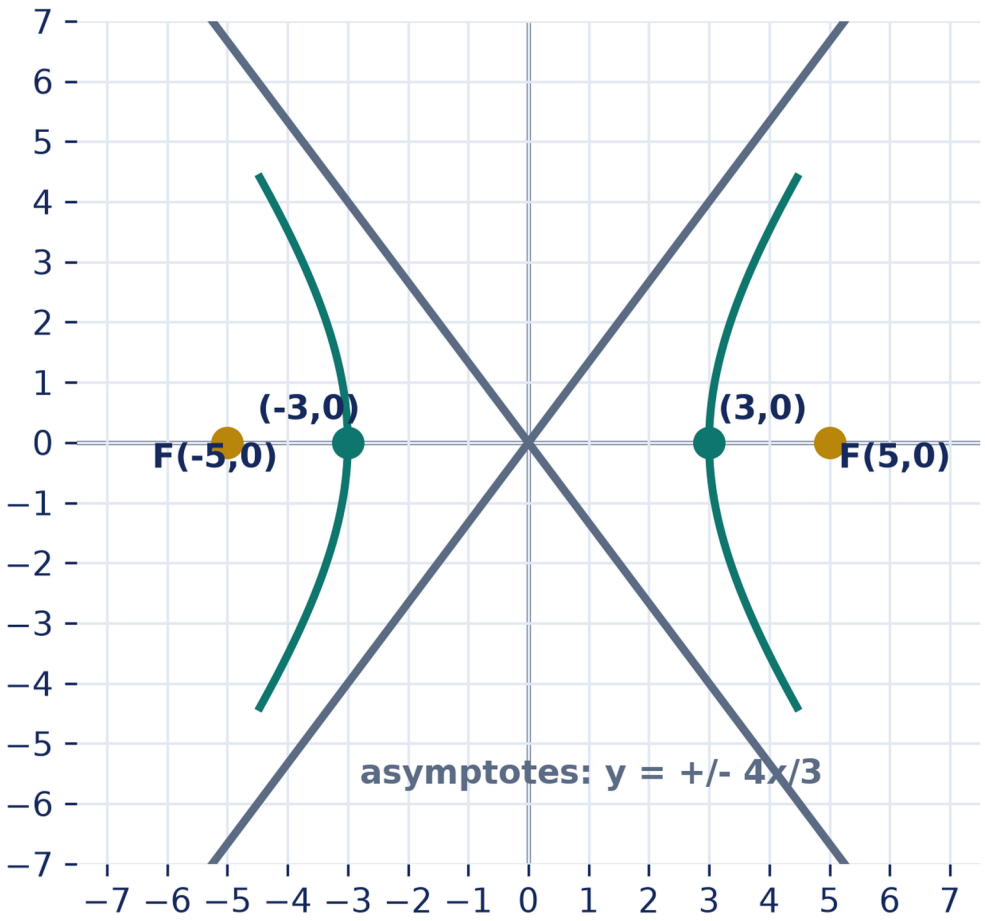
$$y = \pm \frac{b}{a} x.$$

These lines are not tangents at finite points. If 2α is the asymptote angle around the transverse axis, then $\tan\alpha = b/a$ and

$$\cos(2\alpha) = \frac{2}{e^2} - 1.$$

A rectangular hyperbola has perpendicular asymptotes, equivalently $a = b$ and $e = \sqrt{2}$ in horizontal standard form.

The 3-4-5 hyperbola and its asymptotes



READ THE FIGURE

The latus rectum is 2b^2/a = 32/3. The directrices lie between the centre and vertices, while the foci lie outside them.

WORKED EXAMPLE

From a distance difference to the full anatomy

Suppose the absolute difference of distances from $(\pm 5, 0)$ is 6. Then $2a = 6$, so $a = 3$, and $c = 5$. Hence $b^2 = 25 - 9 = 16$ and

$$\frac{x^2}{9} - \frac{y^2}{16} = 1, \qquad e = \frac{5}{3}, \qquad x = \pm\frac{9}{5}$$

are the hyperbola, eccentricity and directrices. Its asymptotes are $y = \pm 4x/3$.

The conjugate shares asymptotes but swaps axes

The conjugate of $x^2/a^2 - y^2/b^2 = 1$ is

$$\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1.$$

Both have the same asymptotes and $c^2 = a^2 + b^2$. If their eccentricities are $e = c/a$ and $e' = c/b$, then

$$\frac{1}{e^2} + \frac{1}{(e')^2} = 1.$$

For $a = 3, b = 4$, this reads $e = 5/3$, $e' = 5/4$, and $9/25 + 16/25 = 1$.

Hyperbola focal chords need branch care

For a **same-branch** focal chord with the focus between its endpoints, the unsigned lengths u, v satisfy

$$\frac{1}{u} + \frac{1}{v} = \frac{2a}{b^2}.$$

For the right focus, write $\ell = b^2/a$. If the two focal radii make opposite angles with the axis, then $u = \ell/(1 - e\cos\theta)$ and $v = \ell/(1 + e\cos\theta)$ in the same-branch case, giving the result. This is not a licence to use the formula for a chord joining opposite branches.

WORKED EXAMPLE

A same-branch check and an opposite-branch warning

For $x^2/9 - y^2/16 = 1$, the right focus is $(5, 0)$. The vertical same-branch focal chord has endpoints $(5, \pm 16/3)$, so $u = v = 16/3$ and

$$\frac{1}{u} + \frac{1}{v} = \frac{3}{8} = \frac{2a}{b^2}.$$

In contrast, the transverse-axis chord joins $(-3, 0)$ and $(3, 0)$. Its focal segments are 8 and 2, whose reciprocal sum is $5/8 = 2c/b^2$, not $3/8$. Branch inspection changes the formula.

KEY RESULT

Use position signs with geometric interpretation

For $S = x^2/a^2 - y^2/b^2 - 1$, $S_1 > 0$ marks the regions containing the two branches, while $S_1 < 0$ marks the region between them. Unlike an ellipse, the plane is not simply described as one inside and one outside region.

LESSON 07 / 10

Tangency conditions carry slope restrictions

At (x_1, y_1) on $x^2/a^2 - y^2/b^2 = 1$,

$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1.$$

Using $P(\theta) = (a\sec\theta, b\tan\theta)$ gives

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1.$$

For a finite-slope tangent,

$$y = mx \pm \sqrt{a^2m^2 - b^2}, \qquad |m| > \frac{b}{a}.$$

The radicand itself explains the restriction: a tangent cannot be as flat as an asymptote direction. Vertex tangents are vertical and come directly from the point form.

The normal at parameter θ can be written

$$ax\sin\theta + by = (a^2 + b^2)\tan\theta.$$

The director locus for perpendicular tangents is $x^2 + y^2 = a^2 - b^2$. It is a real circle only when $a > b$; when $a = b$, a rectangular hyperbola has no real pair of perpendicular tangents, and when $a < b$ the formal radius squared is negative.

WORKED EXAMPLE

A line parallel to an asymptote leaves one finite intersection

Intersect $y = 3x/2 + 1$ with $x^2/4 - y^2/9 = 1$. The line is parallel to the asymptote $y = 3x/2$. Substitution cancels the quadratic terms:

$$-\frac{x}{3} - \frac{1}{9} = 1 \quad\Longrightarrow\quad x = -\frac{10}{3}, \qquad y = -4.$$

There is exactly one finite intersection, $(-10/3, -4)$; the other algebraic intersection has escaped to infinity.

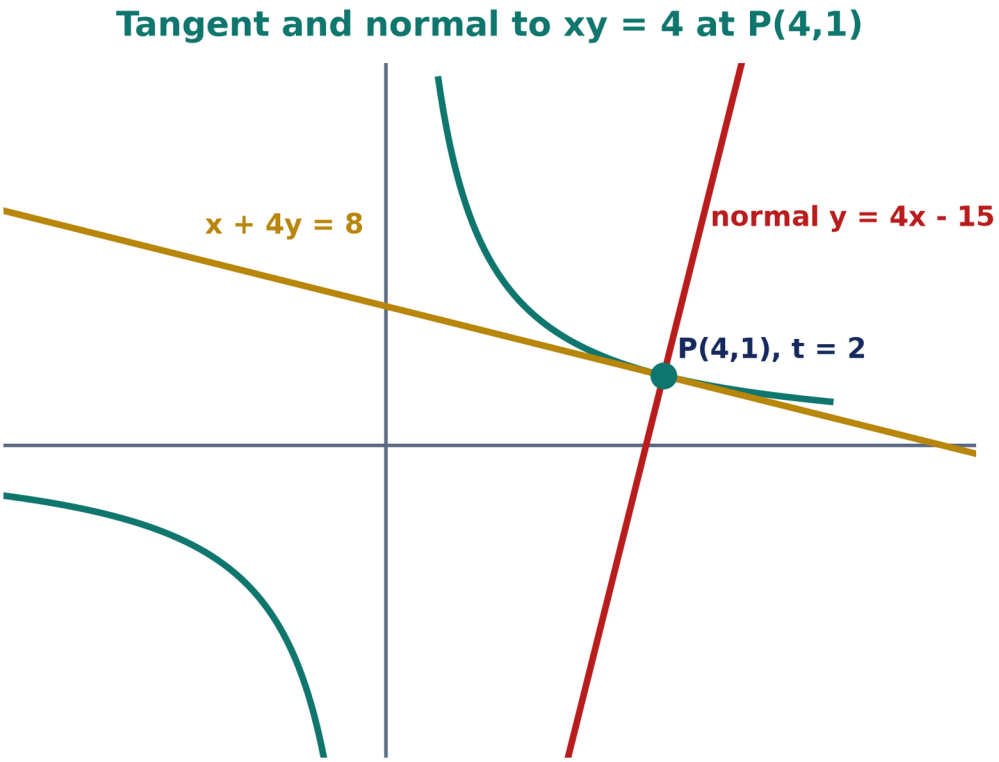
Rotate the rectangular hyperbola into $xy = c^2$

The curve $xy = c^2$ is a rectangular hyperbola with coordinate axes as asymptotes and $e = \sqrt{2}$. Its parameter and tangent/normal pair are

$$P(t) = \left(ct, \frac{c}{t}\right), \qquad \frac{x}{t} + ty = 2c,$$

$$y = t^2x - ct^3 + \frac{c}{t}, \qquad t \neq 0.$$

The tangent slope is $-1/t^2 < 0$, so the normal slope is t^2.



READ THE FIGURE
With c = 2 and t = 2, P = (ct,c/t) = (4,1). The coordinate axes are the asymptotes. The tangent and normal slopes are -1/4 and 4, whose product is -1.

WORKED EXAMPLE

Read both lines from one parameter

For $xy = 4$, $c = 2$. At $t = 2$, the contact point is $(4, 1)$, the tangent is

$$\frac{x}{2} + 2y = 4 \quad \Longleftrightarrow \quad x + 4y = 8,$$

and the normal is $y = 4x - 15$. Substitution of $(4, 1)$ verifies both equations immediately.

COMMON TRAP

Asymptotes are not tangent lines

They describe limiting direction, not finite contact. In particular, a point lying on $y = \pm(b/a)x$ cannot belong to the standard hyperbola. This membership check can prove that requested data are impossible before any lengthy calculation.

LESSON 08 / 10

Define S, S_1 and T once

Write a conic as $S(x,y)=0$. Let $S_1=S(x_1,y_1)$. Obtain T by the symmetric replacements

$$x^2 \mapsto xx_1, \quad y^2 \mapsto yy_1, \quad 2xy \mapsto xy_1+x_1y,$$

$$2x \mapsto x+x_1, \quad 2y \mapsto y+y_1, \quad \text{constants unchanged.}$$

For example, the ellipse has $S=x^2/a^2+y^2/b^2-1$ and $T=xx_1/a^2+yy_1/b^2-1$. For $y^2=4ax$, use $S=y^2-4ax$ and $T=yy_1-2a(x+x_1)$.

KEY RESULT

Four meanings of the same notation

- At an on-conic point, the tangent is $T=0$.
- From an external point, the chord of contact is $T=0$.
- The chord with midpoint (x_1,y_1) is $T=S_1$, provided the resulting chord is real.
- The joint equation of the pair of tangents from (x_1,y_1) is $SS_1=T^2$.

These formulas use consistent conventions. Mixing a conic normalized with -1 against a T expression without that -1 destroys the result.

Chord of contact: two tangents, one line

If tangents from (x_1,y_1) touch at Q and R, each contact point satisfies the condition for its tangent to pass through (x_1,y_1). That condition is precisely $T=0$, so both contacts lie on this one line.

WORKED EXAMPLE

Contacts from $(4,2)$ to $x^2/16+y^2/4=1$

The chord of contact is

$$\frac{4x}{16}+\frac{2y}{4}=1 \iff x+2y=4.$$

Intersecting it with the ellipse gives $Q=(4,0)$ and $R=(0,2)$. The tangents are respectively $x=4$ and $y=2$, and both contain $(4,2)$. This verifies both the contact points and the chord equation.

A midpoint condition is a Vieta condition

For a chord through (x_1,y_1) in direction (u,v), substituting (x_1+tu,y_1+tv) gives a quadratic in t. The midpoint is exactly the supplied point when the two roots are opposite, so their sum is zero. Removing that linear coefficient yields $T=S_1$.

WORKED EXAMPLE

An ellipse chord bisected at $(2, 1)$

For $x^2/16 + y^2/9 = 1$,

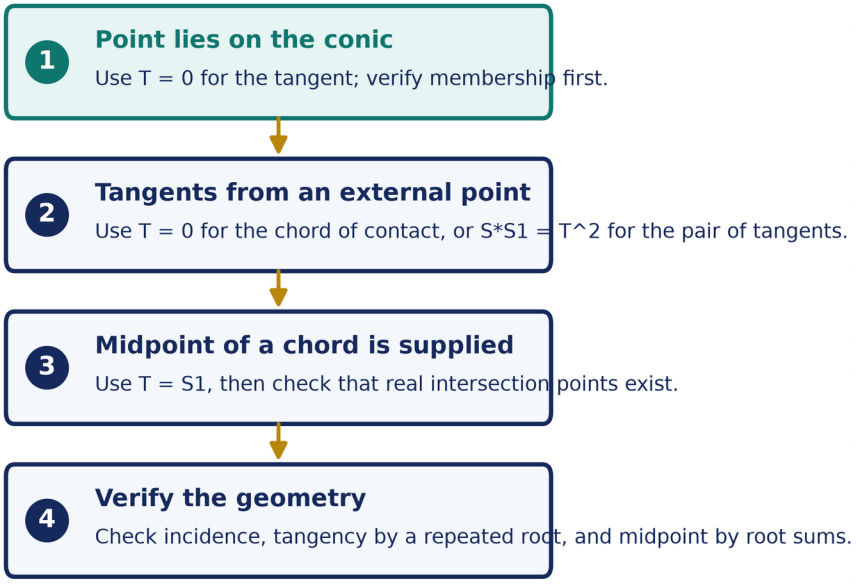
$$S_1 = \frac{4}{16} + \frac{1}{9} - 1 = -\frac{23}{36}.$$

Thus $T = S_1$ becomes

$$\frac{2x}{16} + \frac{y}{9} = \frac{4}{16} + \frac{1}{9} \quad \Longleftrightarrow \quad 9x + 8y = 26.$$

Substitution into the ellipse gives a quadratic whose two x-roots sum to 4 and whose discriminant is positive; the corresponding y-roots sum to 2. Hence this is a genuine chord, not merely a formal line.

Choose the unified conic tool by the wording of the question



READ THE FIGURE

The formula is only the start. A final incidence or reality check prevents a formally correct line from describing an impossible configuration.

COMMON TRAP

A midpoint on the conic is a boundary case

If (x_1, y_1) lies on the conic, the two chord endpoints must coalesce there for it to be their midpoint. The formula $T = S_1$ then reduces to the tangent $T = 0$ because $S_1 = 0$. Do not expect two distinct endpoints in this case.

LESSON 09 / 10

Polarity turns points into lines

For a point $P=(x_1,y_1)$, the line $T=0$ is its **polar**, and P is the **pole** of that line. If P is external, the polar is the chord of contact. If P lies on the conic, the polar is the tangent at P. The definition remains meaningful for interior points too.

SHORT DERIVATION

La Hire's theorem is symmetry

Polar equations use a symmetric bilinear expression. Therefore

$$Q \text{ lies on the polar of } P \iff P \text{ lies on the polar of } Q.$$

This incidence reciprocity is La Hire's theorem. It often replaces a fresh equation-solving step with one substitution.

Two special cases are worth immediate recognition:

- The polar of a focus is its corresponding directrix. For the ellipse, the polar of $(ae,0)$ is $aex/a^2=1$, hence $x=a/e$.
- The pole of a tangent is its contact point. Conversely, the pole of a chord is the intersection of the tangents at its endpoints.

WORKED EXAMPLE

Find a pole by matching coefficients

For $y^2=4x$, the polar of (x_1,y_1) is $yy_1=2(x+x_1)$. To obtain the vertical line $x=8$, we need $y_1=0$ and $2(x+x_1)$ proportional to $x-8$. Thus $x_1=-8$. The pole is $(-8,0)$; indeed its polar is $0=2(x-8)$.

The three director loci are not interchangeable

The locus of intersections of perpendicular tangents is

$$\text{ellipse: } x^2+y^2=a^2+b^2,$$

$$\text{hyperbola: } x^2+y^2=a^2-b^2,$$

while for a parabola it is the directrix $x=-a$. For the hyperbola, reality requires $a>b$; the rectangular case has no real perpendicular tangent pair.

WORKED EXAMPLE

One ellipse, several coordinated answers

For $x^2/25 + y^2/9 = 1$ and P with eccentric angle 60°:

$$P = \left(\frac{5}{2}, \frac{3\sqrt{3}}{2}\right), \qquad e = \frac{4}{5}.$$

Its focal distances are $a - ex = 3$ and $a + ex = 7$. The tangent at P is

$$\frac{x\cos 60^\circ}{5} + \frac{y\sin 60^\circ}{3} = 1 \iff 3x + 5\sqrt{3}y = 30.$$

The director circle is $x^2 + y^2 = 34$. Each result comes from a different named tool; none requires expanding two distance radicals.

A reliable mixed-conic workflow

1. **Name the curve and orientation**
   Read denominators and signs; identify axes, a, b, c and e before substituting data.
2. **Choose coordinates or parameters**
   Use t for parabola chords, eccentric angle for ellipse, sec/tan for hyperbola, and ct,c/t for xy=c^2.
3. **Match the requested object**
   Use focal-distance laws, tangent forms, T = 0, T = S1, or polarity rather than expanding by default.
4. **Close with an independent check**
   Verify membership, incidence, slope product, discriminant, root sum, or a focal-distance sum/difference.

READ THE FIGURE

Most conic errors are convention errors rather than difficult algebra. A ten-second final check catches a wrong directrix, sign, branch, or parameter immediately.

COMMON TRAP

A formula cannot rescue impossible data

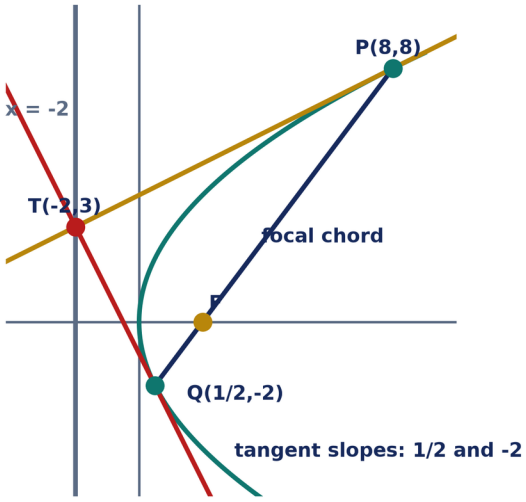
Check that the point lies on the claimed curve, the slope lies in the allowed tangent range, and a requested chord has real endpoints. If the geometry is impossible, the correct response is to report that fact—not to continue until an algebraic expression appears.

LESSON 10 / 10

A focal chord makes its tangents meet on the directrix

For $y^2 = 8x$, the focal chord parameters are $t = 2$ and $t = -1/2$. Its endpoints are $P = (8, 8)$ and $Q = (1/2, -2)$, and it passes through $F = (2, 0)$. The tangents are $2y = x + 8$ and $y = -2x - 1$; they meet at $T = (-2, 3)$ on the directrix and are perpendicular.

Parameters 2 and -1/2 produce a focal chord and perpendicular tangents



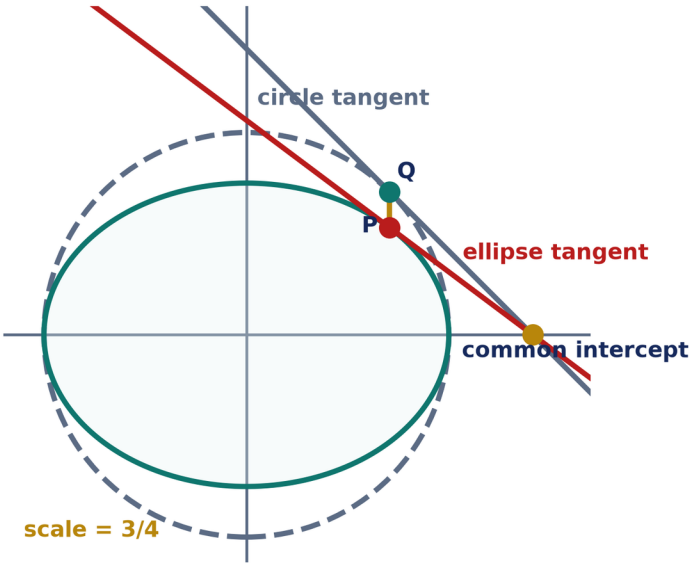
READ THE FIGURE
The tangent slopes are 1/2 and -2. Their product is -1, and the intersection formula gives T = (a*t1*t2,a*(t1+t2)) = (-2,3).

The picture is a checklist: $t_1 t_2 = -1$ gives the focal chord, perpendicular tangents, and directrix intersection together. Without that product, none of these three conclusions should be assumed.

The eccentric angle is seen on the auxiliary circle

For $x^2/16 + y^2/9 = 1$ at $\theta = 45^\circ$, the circle point is $Q = (2\sqrt{2}, 2\sqrt{2})$ and the ellipse point is $P = (2\sqrt{2}, 3/\sqrt{2})$. The circle tangent $x + y = 4\sqrt{2}$ and ellipse tangent $3x + 4y = 12\sqrt{2}$ meet the x-axis at the same point $(4\sqrt{2}, 0)$.

Squashing preserves the common axis intercept of the tangents



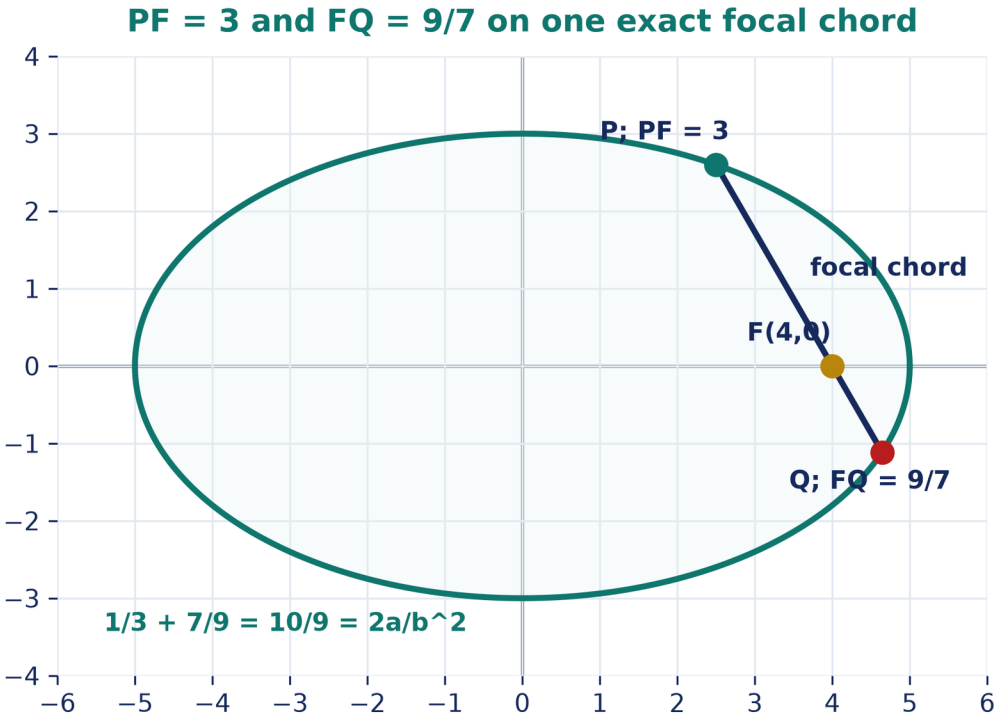
READ THE FIGURE

The eccentric angle belongs to Q on the circle. P has the same x-coordinate but its y-coordinate is multiplied by b/a = 3/4.

When an ellipse problem becomes algebraically heavy, unsquash it to the auxiliary circle, use circle geometry, and then squash back. Keep the distinction between eccentric angle and polar angle explicit.

A focal chord can be checked exactly, not approximately

On $x^2/25 + y^2/9 = 1$, the line through $P = (5/2, 3\sqrt{3}/2)$ and $F = (4, 0)$ meets the ellipse again at $Q = (65/14, -9\sqrt{3}/14)$. The focal segments are $PF = 3$ and $FQ = 9/7$.



READ THE FIGURE

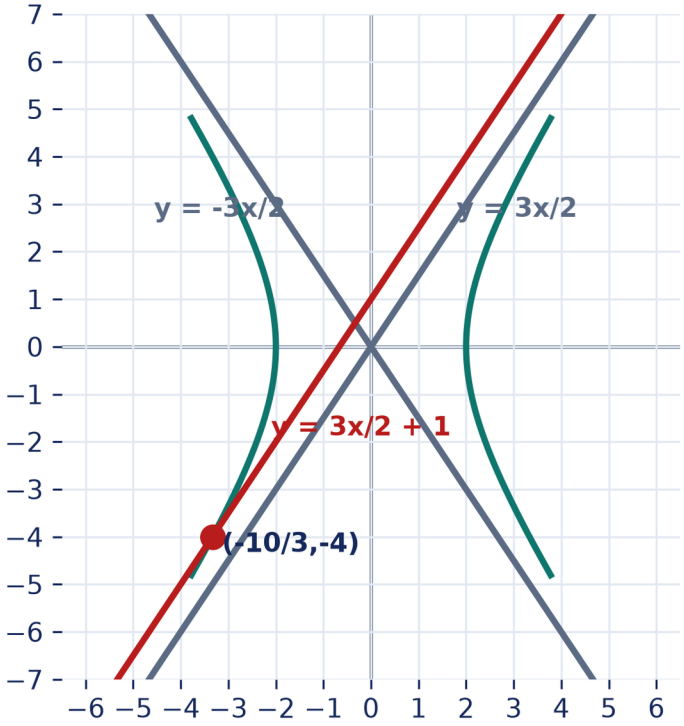
The harmonic mean of the two segment lengths is 9/5 = b^2/a. Q has x = 65/14 < 5 and satisfies the ellipse equation exactly.

The diagram immediately rejects an impossible alleged endpoint beyond $x = 5$. Algebraic exactness and geometric membership should reinforce each other.

Parallel to an asymptote means one finite intersection

For $x^2/4 - y^2/9 = 1$, the line $y = 3x/2 + 1$ is parallel to the asymptote $y = 3x/2$. Substitution becomes linear and gives the single finite intersection $(-10/3, -4)$.

The quadratic collapses along an asymptote direction



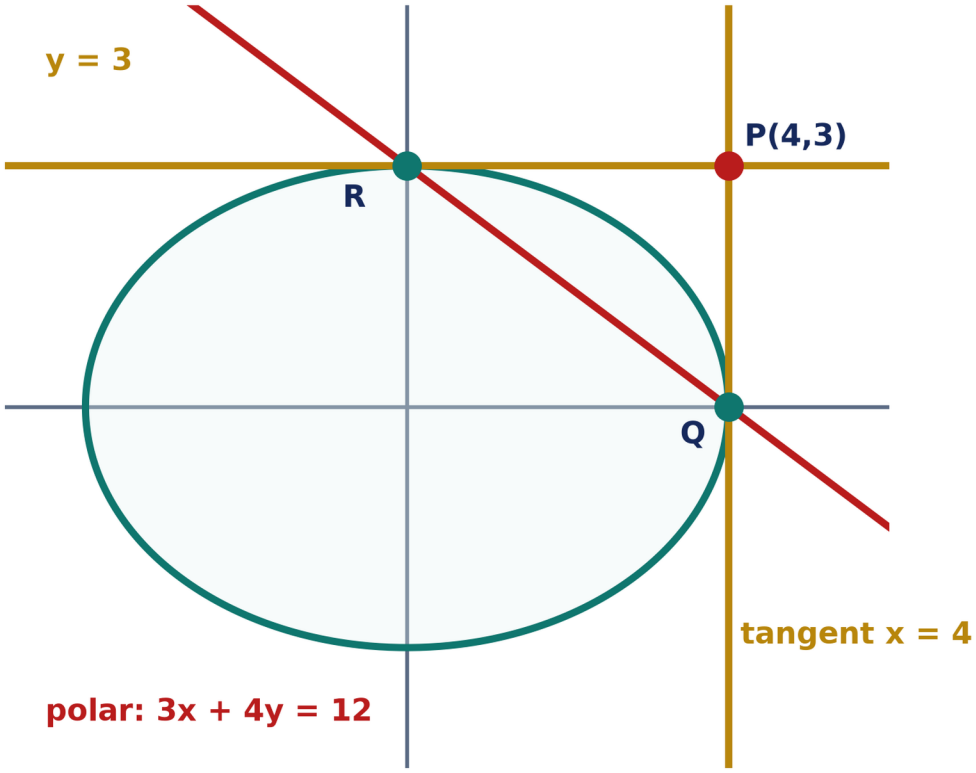
READ THE FIGURE
The curve never meets the asymptote itself. A distinct parallel line has one finite intersection; the second intersection is at infinity.

This is why discriminant formulas must be applied after checking the degree of the substituted equation. If the quadratic coefficient vanishes, the equation is linear and the usual two-root language no longer applies.

Pole and polar preserve incidence

For the ellipse $x^2/16 + y^2/9 = 1$, the polar of $P = (4, 3)$ is $x/4 + y/3 = 1$. It meets the ellipse at $Q = (4, 0)$ and $R = (0, 3)$. The tangents at those contacts are $x = 4$ and $y = 3$, and both pass through P.

The chord of contact joins Q(4,0) and R(0,3)



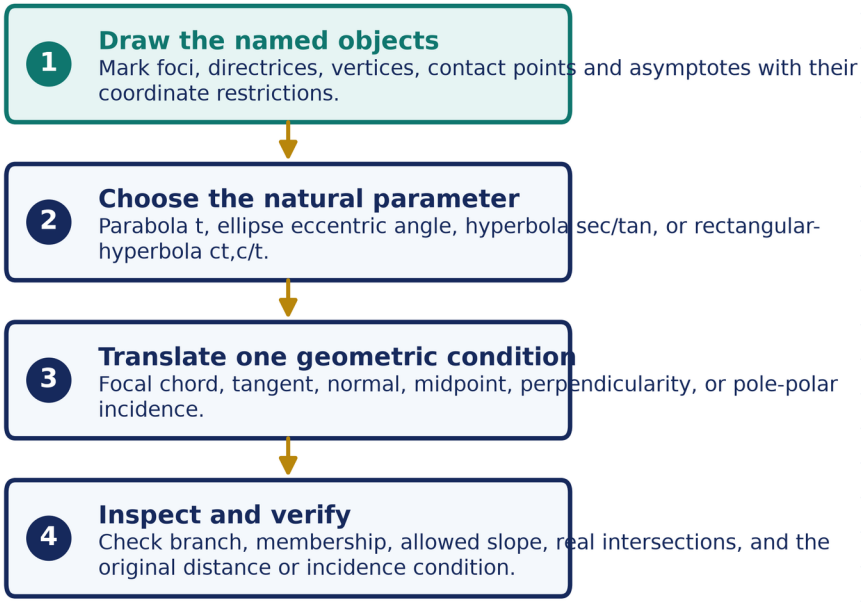
READ THE FIGURE

Q and R lie on the polar of P. La Hire's theorem says that P must lie on the polar of Q and on the polar of R; those polars are the two displayed tangents.

Polarity is not a new computational trick so much as a way to read tangency and incidence backwards. Use it when a problem asks for a pole, a chord of contact, or a concurrent family of polars.

Visual conic strategy

Let the geometry choose the shortest reliable computation



READ THE FIGURE

A sketch is not a substitute for algebra, but it is an excellent error detector. Use both: geometry to choose and check, algebra to compute exactly.

KEY RESULT

The final ten-second check

Ask: Is the point on the curve? Is the focus on the correct side? Does the directrix have the right coordinate? Are tangent slopes allowed? Does a focal chord pass through the focus? These questions catch most conic mistakes before an answer is written.

LESSON 01 / 03

The focal properties are geometric angle laws

A parabola sends rays parallel to its axis through the focus. At an ellipse point, the tangent makes equal angles with the two focal segments, so a ray from one focus reflects toward the other. For a hyperbola, a directed ray toward one focus reflects so that its line passes through the other focus; state the directed-ray convention carefully because a hyperbola has two branches.

A convenient unifying statement uses the normal: at a central conic, the normal makes equal angles with the lines joining the point to the two foci. For the parabola, the normal bisects the angle between the focal segment and a line parallel to the axis.

SHORT DERIVATION

Why the ellipse reflects between its foci

Along a line $PF_1 + PF_2$ changes by the sum of two projected displacements. On a tangent to the level curve $PF_1 + PF_2 = 2a$, the total first-order change is zero. Hence the two unit focal directions have opposite tangential projections, which is exactly the equal-angle law. The same level-set idea gives the directed hyperbola version.

Confocal means shared foci, not merely shared centre

An ellipse and hyperbola are confocal when their foci coincide. In centred horizontal form,

$$\frac{x^2}{A} + \frac{y^2}{B} = 1, \qquad A > B > 0,$$

has $c^2 = A - B$, while

$$\frac{x^2}{C} - \frac{y^2}{D} = 1$$

has $c^2 = C + D$. Confocality therefore requires $A - B = C + D$. Two confocal central conics through a common point have orthogonal tangents there.

WORKED EXAMPLE

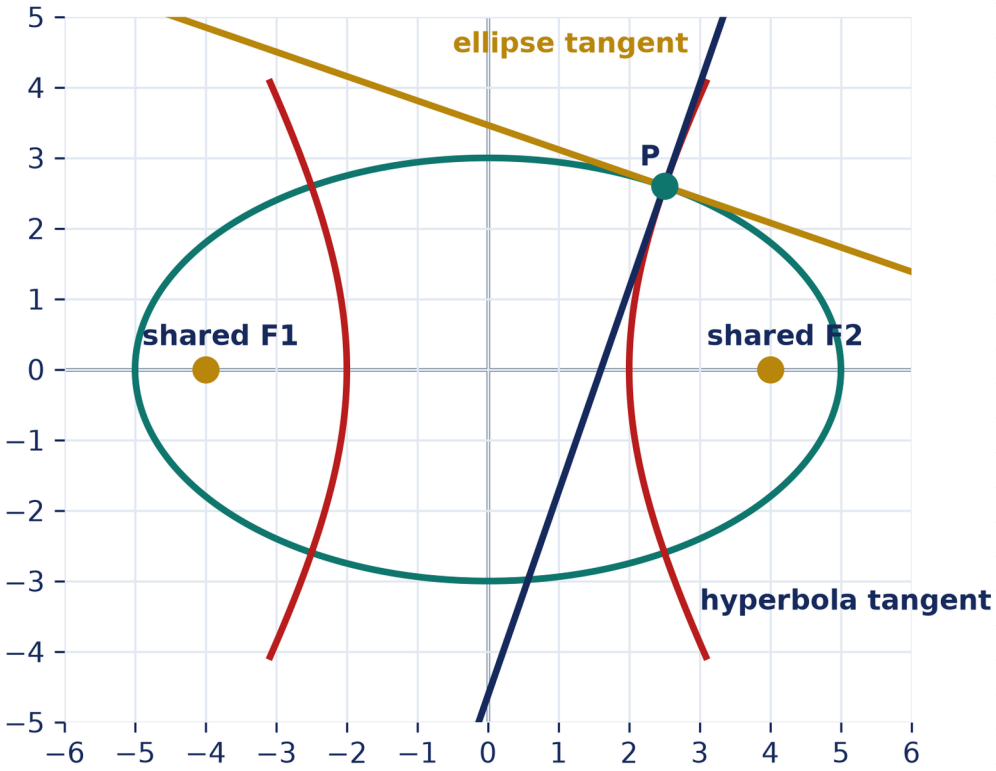
An exact orthogonal confocal intersection

The ellipse $x^2/25 + y^2/9 = 1$ and hyperbola $x^2/4 - y^2/12 = 1$ both have foci $(\pm 4, 0)$ because $25 - 9 = 4 + 12 = 16$. They meet at

$$P = \left(\frac{5}{2}, \frac{3\sqrt{3}}{2}\right).$$

The tangent slopes at P are respectively $-\sqrt{3}/5$ and $5/\sqrt{3}$, whose product is -1. Thus the two curves and their tangent lines are orthogonal at P.

Confocal ellipse and hyperbola meet orthogonally



READ THE FIGURE

The ellipse is x^2/25 + y^2/9 = 1 and the hyperbola is x^2/4 - y^2/12 = 1. Both use c^2 = 16. The tangent slopes are -sqrt(3)/5 and 5/sqrt(3), whose product is -1.

KEY RESULT

Two useful problem transformations

Reflection problems often become straight-line problems after reflecting a focus across a tangent. Confocal problems often become elliptic-coordinate problems after introducing a parameter λ and studying $x^2/(a^2+\lambda) + y^2/(b^2+\lambda) = 1$. Check the denominator signs to determine whether a member is an ellipse, hyperbola, or degenerate locus.

COMMON TRAP

Do not reverse the reflection direction

An ellipse ray from one focus reaches the other. A hyperbola statement needs directed lines and branch awareness; saying simply that both foci always receive the same physical ray is imprecise. Draw the incoming and reflected directions before applying an optical argument.

LESSON 02 / 03

Four concyclic-style incidences on $xy = c^2$

Take three noncollinear points on a rectangular hyperbola, with distinct nonzero parameters:

$$A = (ct_1, \tfrac{c}{t_1}), \quad B = (ct_2, \tfrac{c}{t_2}), \quad C = (ct_3, \tfrac{c}{t_3}).$$

The slope of AB is $-1/(t_1t_2)$, so an altitude perpendicular to it has slope t_1t_2. Imposing the two altitude conditions gives the orthocentre

$$H = (-\tfrac{c}{t_1t_2t_3}, -ct_1t_2t_3).$$

This point also lies on $xy = c^2$, because its coordinates have product c^2. Thus, for a nondegenerate triangle inscribed in a rectangular hyperbola, **the orthocentre lies on the same hyperbola**.

SHORT DERIVATION

Why the formula works

The altitude through A is perpendicular to BC, whose slope is $-1/(t_2t_3)$; its slope is therefore t_2t_3. The line through H and A has exactly that slope after the parameter formula is substituted. Cyclic permutation verifies the other two altitudes, so all three concur at H.

WORKED EXAMPLE

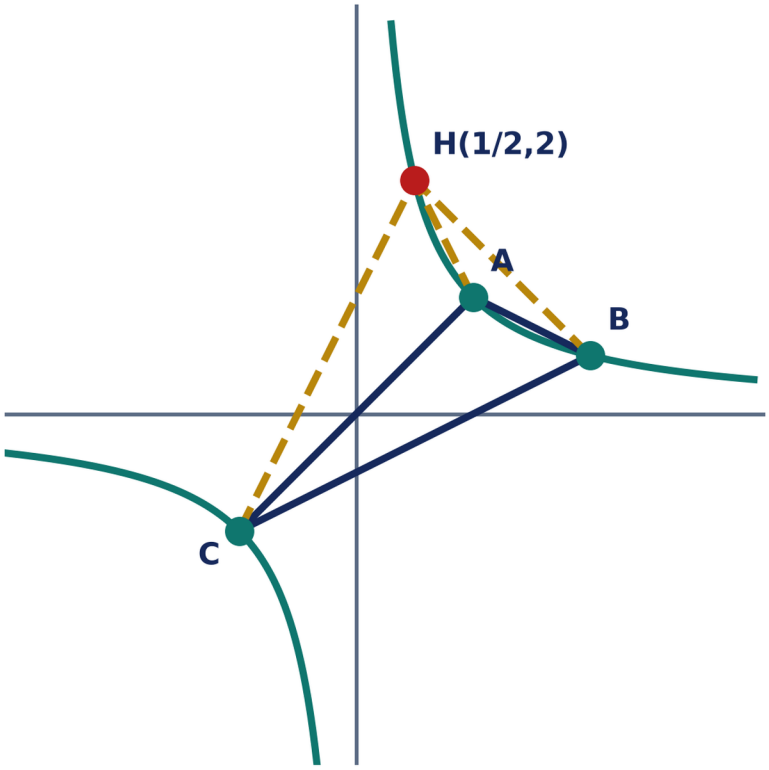
A small exact configuration

On $xy = 1$, take parameters $1, 2, -1$:

$$A = (1, 1), \qquad B = (2, \tfrac{1}{2}), \qquad C = (-1, -1).$$

Their product is -2, so the formula gives $H = (1/2, 2)$. Indeed BC has slope $1/2$, and AH has slope -2; AC has slope 1, and BH has slope -1. All four points lie on $xy = 1$.

A, B, C and their orthocentre H all lie on xy = 1



READ THE FIGURE
The dashed segments lie along the three altitudes. The slope checks AH perpendicular BC and BH perpendicular AC are exact; the third altitude follows by concurrence.

Connect the configuration to the Euler line

For any triangle with circumcentre O, orthocentre H, and nine-point centre N,

$$N = \frac{O+H}{2}, \qquad R_{\text{nine-point}} = \frac{R_{\text{circum}}}{2}.$$

These are general triangle facts, not special hypotheses about the hyperbola. The rectangular-hyperbola theorem adds the powerful incidence $H \in xy = c^2$; combining it with Euler-line geometry often exposes a midpoint, circle, or reflection that coordinates alone would hide.

KEY RESULT
A productive proof pattern
Parameterize three points by t_1, t_2, t_3, compute a slope as $-1/(t_i t_j)$, and use the orthocentre formula when perpendicularity appears. If the problem involves a fourth point on the same rectangular hyperbola, test whether it is the orthocentre of the other three before introducing more variables.

LESSON 03 / 03

Read a general conic through invariants

Write

$$F(x,y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0,$$

and define

$$I = a + b, \qquad \delta = ab - h^2, \qquad \Delta = a(bc - f^2) - h(hc - fg) + g(hf - bg).$$

Here Δ is the determinant of the symmetric coefficient matrix with rows (a,h,g), (h,b,f) and (g,f,c). The quadratic part controls directions at infinity, while Δ detects many degeneracies. For nondegenerate real conics: $\delta > 0$ is elliptic, $\delta = 0$ parabolic, and $\delta < 0$ hyperbolic. In the elliptic case, $\Delta \neq 0$ and $I\Delta < 0$ gives a real ellipse; $I\Delta > 0$ gives no real locus.

COMMON TRAP

Use signs and the coefficient convention together

For example, $x^2 + y^2 + 1 = 0$ has $\delta > 0$ but no real points. When $\Delta = 0$, factor or complete squares to distinguish points, line pairs and empty loci. Preserve the coefficients $2h, 2g, 2f$; nonzero scaling leaves the locus unchanged but scales Δ, so use nonvanishing and consistent sign patterns, never the determinant's numerical size.

General conic decision table for real coefficients

Algebraic signal	Nondegenerate reading	Degenerate caution
delta > 0, Delta != 0	elliptic; use I*Delta for reality	Delta = 0 can collapse to a point
delta = 0, Delta != 0	parabola	Delta = 0 can give parallel/repeated lines or no r
delta < 0, Delta != 0	hyperbola	Delta = 0 can give two intersecting lines
Delta = 0	do not use the nondegenerate rule	factor or complete squares to identify the actua

READ THE FIGURE

These are classification guides, not substitutes for checking real points and degeneracy. The notation Delta is the 3 by 3 determinant and delta = ab - h^2.

Rotation removes the cross term

A rotation of axes through an angle θ satisfying

$$\tan(2\theta) = \frac{2h}{a-b}$$

removes the xy term; when $a = b \neq 0$, use $\theta = \pi/4$. Translations then complete the squares for central conics. Orthogonal changes preserve lengths and angles, so they are preferable when the geometry itself matters.

WORKED EXAMPLE

Classify and rotate in one example

For

$$F = x^2 + 4xy + y^2 - 1 = 0,$$

we have $a = b = 1, h = 2, c = -1$, so $\delta = 1 - 4 = -3$ and $\Delta = 3 \neq 0$. It is a hyperbola. Set $x = (X - Y)/\sqrt{2}$ and $y = (X + Y)/\sqrt{2}$; then

$$F = 3X^2 - Y^2 - 1.$$

Thus the rotated standard form is $3X^2 - Y^2 = 1$, whose asymptotes are $Y = \pm\sqrt{3}X$ in the rotated coordinates.

Degeneracy is exact algebra, not a drawing judgement

If the full expression factors into real linear factors, the locus is the union of their lines. For instance,

$$4x^2 - 12xy + 9y^2 = (2x - 3y)^2 = 0$$

is one repeated line, while $x^2 - y^2 = (x - y)(x + y)$ is two intersecting lines. Both have $\Delta = 0$. Completing squares can also expose a point or an empty real locus.

Pascal and Brianchon: projective companions

For a nondegenerate conic, **Pascal's theorem** says that for a hexagon inscribed in the conic, the intersections of the three pairs of opposite sides are collinear. **Brianchon's theorem** is dual: for a hexagon whose sides are tangent to a conic, the three principal diagonals are concurrent. Degenerate conic versions give classical line theorems such as Pappus.

KEY RESULT

When these tools help

Use invariants for classification without heavy rotation, and Pascal/Brianchon for incidence problems with many intersections or tangents. They are projective statements: keep track of the six-point or six-line ordering, and allow parallel-line intersections to be interpreted at infinity when needed.